

Supplementary Materials for Locus of Ownership and Algorithm Valuations

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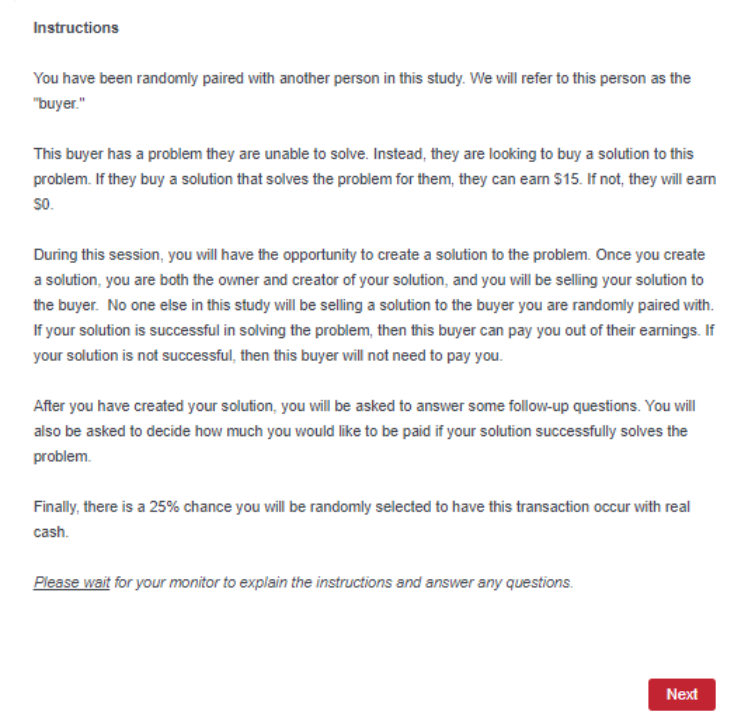
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1 Baseline Study

This section contains screenshots of the computerized survey (hosted with Qualtrics) delivered to participants recruited to participate in the study in a physical laboratory. Non-developers were recruited and paid for approximately 30 minute sessions. Developers were recruited and paid for approximately 40 minute sessions. The nature of the roles was not revealed to subjects in the study’s recruitment ads and exogenously determined; however, developer sessions were separate from nondeveloper sessions to prevent contamination among subjects since the different duration and nature (e.g., use of scratch paper to create versus evaluate an algorithm) of the task could be visible to other persons in the room. The sessions were conducted at a U.S. university’s research laboratory with a maximum capacity of 24 seats per session. Subjects were paid in cash denominations.

1.1 Instructions

Figure 1: Developer Instructions



Instructions

You have been randomly paired with another person in this study. We will refer to this person as the "buyer."

This buyer has a problem they are unable to solve. Instead, they are looking to buy a solution to this problem. If they buy a solution that solves the problem for them, they can earn \$15. If not, they will earn \$0.

During this session, you will have the opportunity to create a solution to the problem. Once you create a solution, you are both the owner and creator of your solution, and you will be selling your solution to the buyer. No one else in this study will be selling a solution to the buyer you are randomly paired with. If your solution is successful in solving the problem, then this buyer can pay you out of their earnings. If your solution is not successful, then this buyer will not need to pay you.

After you have created your solution, you will be asked to answer some follow-up questions. You will also be asked to decide how much you would like to be paid if your solution successfully solves the problem.

Finally, there is a 25% chance you will be randomly selected to have this transaction occur with real cash.

Please wait for your monitor to explain the instructions and answer any questions.

Next

Figure 2: Provider Instructions

Instructions

You have been randomly paired with another person in this study. We will refer to this person as the "buyer."

This buyer has a problem they are unable to solve. Instead, they are looking to buy a solution to this problem. If they buy a solution that solves the problem for them, they can earn \$15. If not, they will earn \$0.

You have been given a solution to the problem that was created by another participant in this study. During this session, you will have a chance to inspect and evaluate your solution. You are the owner of your solution, and you will be selling your solution to the buyer. No one else in this study will be selling a solution to the buyer you are randomly paired with. If your solution is successful in solving the problem, then this buyer can pay you out of their earnings. If your solution is not successful, then this buyer will not need to pay you.

After you have inspected and evaluated your solution, you will be asked to answer some follow-up questions. You will also be asked to decide how much you would like to be paid if your solution successfully solves the problem.

Finally, there is a 25% chance you will be randomly selected to have this transaction occur with real cash.

Please wait for your monitor to explain the instructions and answer any questions.

Next

Figure 3: User Instructions

Instructions

You have been randomly paired with another person in this study. We will refer to this person as the "seller."

There is a problem you will not have the opportunity to solve in this session. Instead, your seller would like to sell you a solution to this problem. If you buy a solution that solves the problem, you can earn \$15. If not, you will earn \$0.

You will be given a chance to look at the solution to the problem that was created by another participant in this study, which the seller would like to sell to you. During this session, you will have a chance to inspect and evaluate this solution. You are *not* the owner of this solution, but you will be allowed to license this solution from your seller. No one else in this study will be buying a solution from the seller you are randomly paired with. If their solution is successful in solving the problem, then you can pay this seller out of your earnings. If their solution is not successful, then you will not need to pay your seller.

After you have inspected and evaluated their solution, you will be asked to answer some follow-up questions. You will also be asked to decide how much you would like to pay if it successfully solves the problem.

Finally, there is a 25% chance you will be randomly selected to have this transaction occur with real cash.

Please wait for your monitor to explain the instructions and answer any questions.

Next

1.2 Comprehension

Figure 4: Developer Comprehension Questions

You will now be asked a few comprehension questions to make sure you understand your task in this study.

1. Are you the owner of the solution that has been given to you to look at?

Yes

No

2. What happens after you inspect and evaluate the seller's solution?

I will license the solution from the seller, and I will earn \$15 whether their solution is accurate or not.

I will license the solution from the seller, and I will earn \$15 only if their solution successfully solves the problem.

3. How can you pay for using their solution?

If their solution is successful, then I can pay them out of my earnings. If my solution is not successful, then I will not need to pay the seller.

I pay them whether their solution successfully solves the problem or not.

Next

Figure 5: Provider Comprehension Questions

You will now be asked a few comprehension questions to make sure you understand your task in this study.

1. Are you the owner of the solution that has been given to you?

Yes

No

2. What happens after you inspect and evaluate your solution?

I will sell my solution to the buyer, and the buyer will earn \$15 whether my solution is accurate or not.

I will sell my solution to the buyer, and the buyer will earn \$15 only if my solution successfully solves the problem.

3. How does the buyer pay you for your solution?

If my solution is successful, then the buyer can pay me at most \$15. If my solution is not successful, then the buyer will not need to pay me.

Whether my solution successfully solves the problem or not, the buyer pays me at most \$15.

Next

Figure 6: User Comprehension Questions

You will now be asked a few comprehension questions to make sure you understand your task in this study.

1. Are you the owner of the solution that has been given to you to look at?

Yes

No

2. What happens after you inspect and evaluate the seller's solution?

I will license the solution from the seller, and I will earn \$15 whether their solution is accurate or not.

I will license the solution from the seller, and I will earn \$15 only if their solution successfully solves the problem.

3. How can you pay for using their solution?

If their solution is successful, then I can pay them out of my earnings. If my solution is not successful, then I will not need to pay the seller.

I pay them whether their solution successfully solves the problem or not.

Next

Figure 7: Developer Comprehension Questions Example Review

Please review your answers to the comprehension questions.

1. After you have come up with a solution to the problem, are you both the creator and owner of this solution?

Your answer: No
Incorrect. You are both the creator and owner of your solution.

2. What happens after you create your solution?

Your answer: I will sell my solution to the buyer, and the buyer will earn \$15 only if my solution successfully solves the problem.
Correct!

3. How does the buyer pay you for your solution?

Your answer: Whether my solution successfully solves the problem or not, the buyer pays me at most \$15.
Incorrect. If your solution is successful, then buyer can pay you at most \$15. If your solution is not successful, then the buyer will not need to pay you.

Next

Figure 8: Provider Comprehension Questions Example Review

Please review your answers to the comprehension questions.

1. Are you the owner of the solution that has been given to you?

Your answer: Yes
Correct!

2. What happens after you inspect and evaluate your solution?

Your answer: I will sell my solution to the buyer, and the buyer will earn \$15 whether my solution is accurate or not.
Incorrect. They will earn money only if your solution is successful.

3. How does the buyer pay you for your solution?

Your answer: If my solution is successful, then the buyer can pay me at most \$15. If my solution is not successful, then the buyer will not need to pay me.
Correct!

Next

Figure 9: User Comprehension Questions Example Review

Please review your answers to the comprehension questions.

1. Are you the owner of the solution that has been given to you to look at?

Your answer: Yes

Incorrect. You are not the owner of the solution that you will be looking at.

2. What happens after you inspect and evaluate the seller's solution?

Your answer: I will license the solution from the seller, and I will earn \$15 whether their solution is accurate or not.

Incorrect. You will earn money only if their solution is successful.

3. How can you pay for using their solution?

Your answer: I can pay them whether their solution successfully solves the problem or not.

Incorrect. If their solution is successful, then you can pay them out of your earnings. If their solution is not successful, then you will not need to pay the seller.

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1.3 Algorithm Task

Figure 10: Developer Algorithm Task

The Problem

There are 6 bars, named A, B, C, D, E, and F. One bar is made of fake gold, and 5 bars are made of real gold. The fake gold bar weighs either heavier or lighter than a real gold bar. All the real gold bars weigh the same.



There is a balancing scale that can weigh these bars. You can place as many bars as you would like on either side of the scale to determine whether one side weighs heavier, lighter, or the same as the other side. However, this scale can ONLY be used up to 3 times.



In the box below, please describe your method for finding the fake gold bar and whether it is heavier or lighter than a real gold bar.

Please write in English or any clearly labeled notation to describe your solution. Once you are satisfied with your solution, you may move onto the next page. If your 30 minute session is over, your solution will be saved as is, and you will be taken to the next page.

Your solution:

29:54

Next

Note: There was a 30 minute limit on the time spent on this page, and the timer is displayed to at the bottom of the page.

Figure 11: Provider Algorithm Task

The Problem

There are 6 bars, named A, B, C, D, E, and F. One bar is made of fake gold, and 5 bars are made of real gold. The fake gold bar weighs either heavier or lighter than a real gold bar. All the real gold bars weigh the same.



There is a balancing scale that can weigh these bars. You can place as many bars as you would like on either side of the scale to determine whether one side weighs heavier, lighter, or the same as the other side. However, this scale can ONLY be used up to 3 times.



Your solution was created by another participant in this study that describes a method for finding the fake gold bar and whether it is heavier or lighter than a real gold bar. Please take this time to carefully assess the accuracy and quality of your solution.

Once you are ready, you may move onto the next page. If your 20 minute session is over, you will be taken to the next page.

1957

Next

Note: There was a 20 minute limit on the time spent on this page, and the timer is displayed to at the bottom of the page.

Figure 12: User Algorithm Task

The Problem

There are 6 bars, named A, B, C, D, E, and F. One bar is made of fake gold, and 5 bars are made of real gold. The fake gold bar weighs either heavier or lighter than a real gold bar. All the real gold bars weigh the same.



There is a balancing scale that can weigh these bars. You can place as many bars as you would like on either side of the scale to determine whether one side weighs heavier, lighter, or the same as the other side. However, this scale can ONLY be used up to 3 times.



The solution that has been given to you to look at was created by another participant (who may not be the same person as your seller, who is the owner of this solution). It describes a method for finding the fake gold bar and whether it is heavier or lighter than a real gold bar. Please take this time to carefully assess the accuracy and quality of your seller's solution.

Once you are ready, you may move onto the next page. If your 20 minute session is over, you will be taken to the next page.

1958

Next

Note: There was a 20 minute limit on the time spent on this page, and the timer is displayed to at the bottom of the page.

Figure 13: Developer End of Algorithm Task

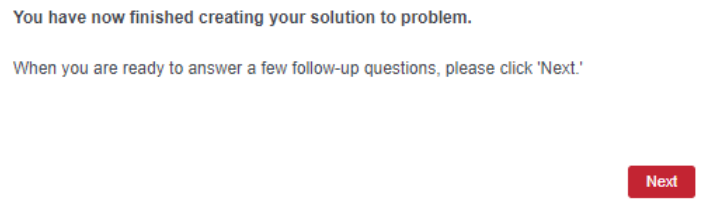


Figure 14: Provider End of Algorithm Task

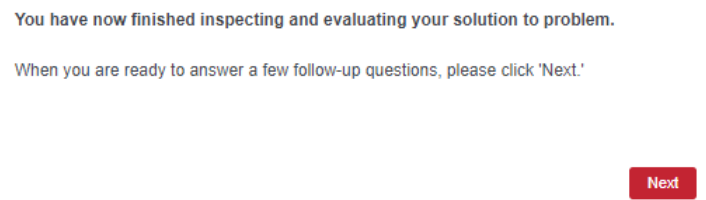
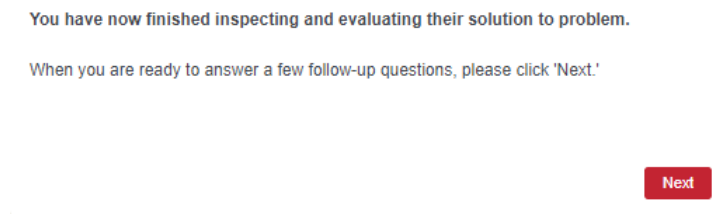


Figure 15: User End of Algorithm Task



1.4 Beliefs

Figure 16: Developer Beliefs

To the best of your knowledge, what is the maximum number of times your solution requires using the scale?

1
2
3
More than 3
Not sure

How confident are you that your solution will accurately solve the problem? An accurate solution is one that can determine the fake gold bar and whether it is heavier or lighter than a real gold bar by using the scale at most 3 times.

Please rank this with a number between 1 and 10 (1 being the lowest amount of confidence, and 10 being the highest amount of confidence).

1	2	3	4	5	6	7	8	9	10
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How would you rank the quality of your solution?

Please rank this with a number between 1 and 10 (1 being the lowest quality, and 10 being the highest quality).

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

Next

Note: Instructions were provided in each experimental session that defined the term “quality” as being independent of algorithmic performance, including writing style and readability.

Figure 17: Provider Beliefs

To the best of your knowledge, what is the maximum number of times your solution requires using the scale?

1
2
3
More than 3
Not sure

How confident are you that your solution will accurately solve the problem? An accurate solution is one that can determine the fake gold bar and whether it is heavier or lighter than a real gold bar by using the scale at most 3 times.

Please rank this with a number between 1 and 10 (1 being the lowest amount of confidence, and 10 being the highest amount of confidence).

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

How would you rank the quality of your solution?

Please rank this with a number between 1 and 10 (1 being the lowest quality, and 10 being the highest quality).

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

Next

Note: Instructions were provided in each experimental session that defined the term “quality” as being independent of algorithmic performance, including writing style and readability.

Figure 18: User Beliefs

To the best of your knowledge, what is the maximum number of times their solution requires using the scale?

1
2
3
More than 3
Not sure

How confident are you that their solution will accurately solve the problem? An accurate solution is one that can determine the fake gold bar and whether it is heavier or lighter than a real gold bar by using the scale at most 3 times.

Please rank this with a number between 1 and 10 (1 being the lowest amount of confidence, and 10 being the highest amount of confidence).

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

How would you rank the quality of their solution?

Please rank this with a number between 1 and 10 (1 being the lowest quality, and 10 being the highest quality).

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

Next

Note: Instructions were provided in each experimental session that defined the term “quality” as being independent of algorithmic performance, including writing style and readability.

1.5 Prices

Figure 19: Developer Price Instructions

As previously mentioned, if your solution is successful in solving the problem, then the buyer can earn \$15.

You will be asked a series of questions about how much you would like to sell your solution for when your buyer can pay you at most \$15. That is, your buyer will only pay you if your solution is successful. If your solution does not solve the problem, then the buyer will not need to pay you.

When you are ready, please click 'Next.'

Next

Figure 20: Provider Price Instructions

As previously mentioned, if your solution is successful in solving the problem, then the buyer can earn \$15.

You will be asked a series of questions about how much you would like to sell your solution for when your buyer can pay you at most \$15. That is, your buyer will only pay you if your solution is successful. If your solution does not solve the problem, then the buyer will not need to pay you.

When you are ready, please click 'Next.'

Next

Figure 21: User Price Instructions

As previously mentioned, if their solution is successful in solving the problem, then you can earn \$15.

You will be asked a series of questions about how much you would like to pay for using (i.e. licensing) their solution. You will only pay this licensing fee if their solution is successful. If their solution does not solve the problem, then you will not need to pay them a licensing fee.

When you are ready, please click 'Next.'

Next

Figure 22: Developer Sample Price Question

Would you sell your solution for **\$15**?

Yes	No
-----	----

Next

Figure 23: Provider Sample Price Question

Would you sell your solution for **\$13**?

Yes	No
-----	----

Next

Figure 24: User Sample Price Question

Would you pay **\$1.00**?

Yes	No
-----	----

Next

Figure 25: Developer Sample Price Choice

You have decided the payment you are willing to accept is **greater than \$9.50**. Is this correct?

Yes, this is correct. I am ready to continue.

No, I would like to change my answer to the previous question. I am willing to accept \$9.50.

Next

Figure 26: Provider Sample Price Choice

You have decided the payment you are willing to accept is **greater than \$7**. Is this correct?

Yes, this is correct. I am ready to continue.

No, I would like to change my answer to the previous question. I am willing to accept \$7.

Next

Figure 27: User Sample Price Choice.

You have decided you are willing to pay **less than \$4.00**. Is this correct?

Yes, this is correct. I am ready to continue.

No, I would like to change my answer to the previous question. I am willing to pay \$4.00.

Next

1.6 Exit Lottery

Figure 28: Exit Lottery for All Subjects

As previously mentioned, there is a 25% chance you will be randomly selected to earn real cash in this transaction.

As part of this random selection, please choose a number:

1

2

3

4

Next

2 Belief-Controlled Study

This section contains screenshots of the computerized survey (hosted with Qualtrics) delivered to panelists recruited to participate in the study online. All participants were non-developers and randomly assigned within the survey to the role of licensor or licensee. Panelists were recruited from the study pool of a U.S. university and paid for a 10 minute online survey. A lottery for bonus rewards were given for completion of the survey. Subjects were paid through Amazon gift cards.

2.1 Instructions

Figure 29: Provider Instructions with Metainformation on Algorithm Quality – 1

Consider the following logic problem:

There are 6 bars, named A, B, C, D, E, and F. One bar is made of fake gold, and 5 bars are made of real gold. The fake gold bar weighs either heavier or lighter than a real gold bar. All the real gold bars weigh the same.



There is a balancing scale that can weigh these bars. You can place as many bars as you would like on either side of the scale to determine whether one side weighs heavier, lighter, or the same as the other side. However, this scale can ONLY be used up to 3 times.



In this study, you will **consider three different solutions** to the problem that are **owned by another participant**. An expert has reviewed each solution and evaluated how successful it is at solving the problem. You will have the chance to review these solutions and their grades. For the purpose of this study, consider the expert as someone (or a robot) who is completely accurate and objective in evaluating solutions to this problem.

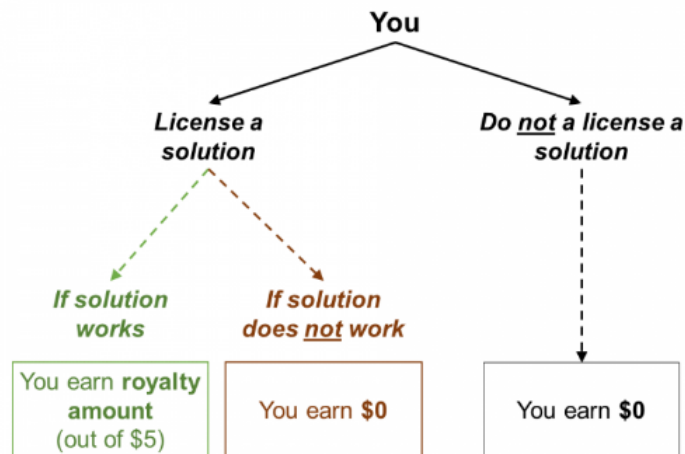
As you consider this owner's solutions, you will have the **ability to license** one if you wish to use it.

Next >

Figure 30: Provider Instructions with Metainformation on Algorithm Quality – 2

There is another participant in this study who has the chance to make \$5 by solving the logic problem. However, they do not own any solutions and they are not allowed to make their own solution. Instead, they can **license one of your solutions** to help solve the problem and make money. If they license one of your solutions and the solution works, they earn \$5 and you earn a royalty amount. If they license one of your solutions and the solution does not work, you earn \$0.

The diagram below shows how licensing one of your solutions can determine your earnings:



You can think of \$5 as the maximum possible value of each of your three solutions. Each of your solutions has some chance to make \$5 for this other participant, but some may be more likely to succeed than others. Later in the survey, you will determine the **minimum royalty amount** (out of \$5) **you are willing to accept** for each of your three solutions.

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Figure 31: Provider Instructions with Metainformation on Algorithm Quality – 3

In the next part, you will be given some comprehension questions about the information you just saw. Please feel free to use the **"Back"** button to review the instructions before continuing.

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Figure 32: User Instructions with Metainformation on Algorithm Quality – 1

Consider the following logic problem:

There are 6 bars, named A, B, C, D, E, and F. One bar is made of fake gold, and 5 bars are made of real gold. The fake gold bar weighs either heavier or lighter than a real gold bar. All the real gold bars weigh the same.



There is a balancing scale that can weigh these bars. You can place as many bars as you would like on either side of the scale to determine whether one side weighs heavier, lighter, or the same as the other side. However, this scale can ONLY be used up to 3 times.



In this study, you will **consider three different solutions** to the problem that are **owned by another participant**. An expert has reviewed each solution and evaluated how successful it is at solving the problem. You will have the chance to review these solutions and their grades. For the purpose of this study, consider the expert as someone (or a robot) who is completely accurate and objective in evaluating solutions to this problem.

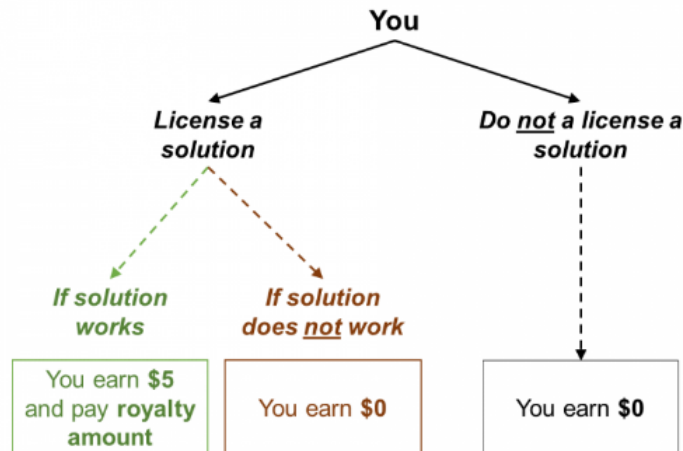
As you consider this owner's solutions, you will have the **ability to license** one if you wish to use it.

Next >

Figure 33: User Instructions with Metainformation on Algorithm Quality – 2

You will have the chance to make \$5 by solving the logic problem. However, you do not own any solutions and are not allowed to make your own solution. Instead, you can **license one of the owner's solutions** to help solve the problem and make money. If you license one of their solutions and the solution works, you earn \$5 and pay a royalty amount. If you license one of their solutions and the solution does not work, you earn \$0.

The diagram below shows how licensing one of the owner's solutions can determine your earnings:



You can think of \$5 as the maximum possible value of each of the owner's three solutions. Each of these solution has some chance to make \$5, but some may be more likely to succeed than others. Later in the survey, you will determine the **maximum royalty amount** (out of \$5) **you are willing to pay** for each of the three solutions.

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Figure 34: User Instructions with Metainformation on Algorithm Quality – 3

In the next part, you will be given some comprehension questions about the information you just saw. Please feel free to use the "**Back**" button to review the instructions before continuing.

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Figure 35: Provider Instructions without Metainformation on Algorithm Quality – 1

Consider the following logic problem:

There are 6 bars, named A, B, C, D, E, and F. One bar is made of fake gold, and 5 bars are made of real gold. The fake gold bar weighs either heavier or lighter than a real gold bar. All the real gold bars weigh the same.



There is a balancing scale that can weigh these bars. You can place as many bars as you would like on either side of the scale to determine whether one side weighs heavier, lighter, or the same as the other side. However, this scale can ONLY be used up to 3 times.



In this study, you will be given **ownership of three different solutions** to this problem. You will review each solution and evaluate how successful it is at solving the problem.

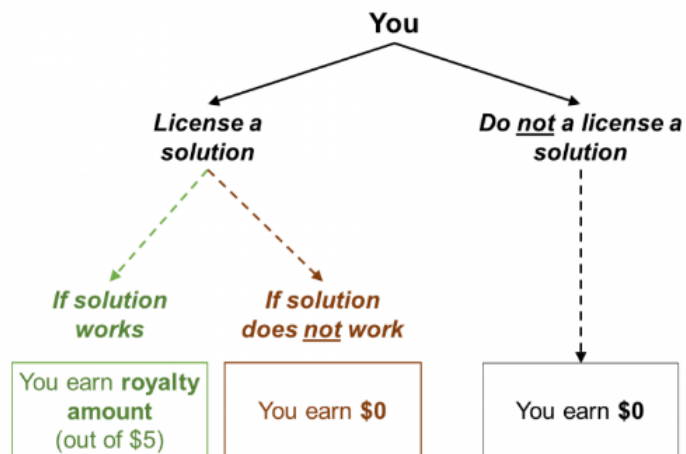
As the owner of these solutions, you have the **ability to license** them to anyone who wishes to use them.

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Figure 36: Provider Instructions without Metainformation on Algorithm Quality – 2

There is another participant in this study who has the chance to make \$5 by solving the logic problem. However, they do not own any solutions and they are not allowed to make their own solution. Instead, they can **license one of your solutions** to help solve the problem and make money. If they license one of your solutions and the solution works, they earn \$5 and you earn a royalty amount. If they license one of your solutions and the solution does not work, you earn \$0.

The diagram below shows how licensing one of your solutions can determine your earnings:



You can think of \$5 as the maximum possible value of each of your three solutions. Each of your solutions has some chance to make \$5 for this other participant, but some may be more likely to succeed than others. Later in the survey, you will determine the **minimum royalty amount** (out of \$5) **you are willing to accept** for each of your three solutions.

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Figure 37: Provider Instructions without Metainformation on Algorithm Quality – 3

In the next part, you will be given some comprehension questions about the information you just saw. Please feel free to use the **"Back"** button to review the instructions before continuing.

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2.2 Comprehension

Figure 38: Provider Comprehension Question 1

In this study, what will you be given ownership of?

☐ I will own three different solutions to the logic problem

☐ I will own one solution to the logic problem

Correct. In this study, you will be given ownership of three different solutions to the logic problem.

Incorrect. In this study, you will be given ownership of not one but three different solutions to the logic problem.

Note: Feedback on whether the respondent's answer was correct or incorrect was provided on the next screen after answer submission.

Figure 39: Provider Comprehension Question 2

Why is there another participant interested in a licensing solution to the logic problem?

☐ They can make money by using a solution to solve the logic problem.

☐ They can make money by selling these solutions to other participants.

Correct. The other participant is interested in licensing a solution since they can make money by solving the logic problem.

Incorrect. The other participant is interested in licensing a solution since they can make money by solving the logic problem.

Note: Feedback on whether the respondent's answer was correct or incorrect was provided on the next screen after answer submission.

Figure 40: Provider Comprehension Question 3

What is the royalty amount?

☐ The amount I earn if I license my solution and it succeeds.

☐ It is the price I earn if I sell a solution regardless of whether it works or not.

Correct. You earn the royalty amount if you license a solution and it succeeds for the other participant. You earn \$0 if you license a solution and it fails for the other participant. If you do not license a solution, you earn \$0.

Incorrect. You earn the royalty amount if you license a solution and it succeeds for the other participant. You earn \$0 if you license a solution and it fails for the other participant. If you do not license a solution, you earn \$0.

Note: Feedback on whether the respondent's answer was correct or incorrect was provided on the next screen after answer submission.

Figure 41: User Comprehension Question 1

How can you solve the logic problem?

☐ I will create my own solution to the logic problem.

☐ I will license a solution owned by another participant in this study.

Correct. In this study, you can solve the logic problem by licensing and using a solution owned by another participant in this study. You will not be given the chance to own or create your own solution.

Incorrect. In this study, you can solve the logic problem by licensing and using a solution owned by another participant in this study. You will not be given the chance to own or create your own solution.

Note: Feedback on whether the respondent's answer was correct or incorrect was provided on the next screen after answer submission.

Figure 42: User Comprehension Question 2

Why are you interested in licensing a solution?

☐ I can make money by selling the solution to others.

☐ I can make money by using a solution to solve the logic problem.

Correct. You can make money by using a solution to solve the logic problem. When you license a solution, you are not given abilities to sell the solution to others for money.

Incorrect. You can make money by using a solution to solve the logic problem. When you license a solution, you are not given abilities to sell the solution to others for money.

Note: Feedback on whether the respondent's answer was correct or incorrect was provided on the next screen after answer submission.

Figure 43: User Comprehension Question 3

What is the royalty amount?

- ☐ It is the price (out of \$5) I pay if I license a solution from the owner and it successfully solves the problem for me.
- ☐ It is the price (out of \$5) I pay if I buy the owner's solution regardless of whether the solution is successful.

Correct. The royalty amount is the price you pay out of your \$5 earnings if you license a solution from the owner and it successfully solves the problem for you. You earn \$0 and do not pay the royalty amount if the solution is not successful.

Incorrect. The royalty amount is the price you pay out of your \$5 earnings if you license a solution from the owner and it successfully solves the problem for you. You earn \$0 and do not pay the royalty amount if the solution is not successful.

Note: Feedback on whether the respondent's answer was correct or incorrect was provided on the next screen after answer submission.

2.3 Algorithms and Prices

Figure 44: Provider Instructions for Price Choice

On the following pages, you will see your three different solutions in a random order. The expert grade for each of your solutions will be displayed above them. At the end of this study, one of your three solutions will be randomly chosen to earn real money for you.

For each solution, you will determine the minimum royalty amount (out of \$5) you are willing to accept to license it. It is in your best interest to answer these questions honestly.

After you determine the minimum you are willing to accept for each solution, the computer will generate a royalty amount between \$0 and \$5. If the minimum you are willing to accept is above the royalty amount, you will not license that solution and earn \$0. If the minimum you are willing to accept is below the royalty amount, you will license that solution. If you license that solution and it works, you will earn the royalty amount. If you license that solution and it does not work, you will earn \$0.

Next >

Figure 45: User Instructions for Price Choice

On the following pages, you will see the three different solutions in a random order. The expert grade for each of these solutions will be displayed above them. At the end of this study, one of these three solutions will be randomly chosen to earn real money for you.

For each solution, you will determine the maximum royalty amount (out of \$5) you are willing to pay to license it. It is in your best interest to answer these questions honestly.

After you determine the maximum you are willing to pay for each solution, the computer will generate a royalty amount between \$0 and \$5. If the maximum you are willing to pay is below the royalty amount, you will not license that solution and earn \$0. If the maximum you are willing to pay is above the royalty amount, you will license that solution. If you license that solution and it works, you will earn \$5 and pay the royalty amount. If you license that solution and it does not work, you will earn \$0 and you will not pay the royalty amount.

Next >

Figure 46: Provider High Quality Algorithm Price Choice

According to the expert, this solution has a **80% or greater chance of success**.

Pair one gold bar with another (no repeats) in order to make 3 separate groups of gold bars with 2 gold bars in each group.

Ex.

Group 1: A and B

Group 2: C and D

Group 3: E and F

Weigh Group 1 against Group 2. You should expect 1 of 2 results. Either the group of bars equally weigh each other or not. Note this result. Then Weigh Group 1 against Group 3. Again, you should expect 1 of 2 results. Either the group of bars should weigh each other or not. Note this result.

Based on your results, this is the solution and follow the description:

If Group 1 and Group 2 did NOT weigh the same AND Group 1 and Group 3 did NOT weigh the same, then you know that the one of the gold bars in Group 1 is fake. Choose either bar A or bar B and then weigh it against one of the gold bars in either Group 2 or Group 3. If the selected bar (let's say Bar B) in Group 1 weighs equally as the selected bar in either Group 2 or Group 3, then you can safely assume that the other bar (Bar A in this case) is the fake gold bar. If the selected bar in Group 1 weighs differently than the selected gold bar in either Group 2 or Group 3, then you can safely assume that the selected bar in Group 1 that you just weighed is the fake gold bar (in this case Bar B).

If Group 1 and Group 2 did NOT weigh the same AND Group 1 and Group 3 did weigh the same, then you know that the one of the gold bars in Group 2 is fake. Choose either bar C or bar D in Group 2 (let's say Bar C) and then weigh it against one of the gold bars in either Group 1 or Group 3. If the selected bar in Group 2 weighs equally as the selected bar in either Group 1 or Group 3, then you can safely assume that the other bar (Bar D in this case) is the fake gold bar. If the selected bar in Group 2 weighs differently than the selected gold bar in either Group 1 or Group 3, then you can safely assume that the selected bar in Group 2 that you just weighed is the fake gold bar (in this case Bar C).

If Group 1 and Group 2 did weigh the same AND Group 1 and Group 3 did NOT weigh the same, then you know that the one of the gold bars in Group 3 is fake. Choose either bar E or bar F in Group 3 (let's say Bar E) and then weigh it against one of the gold bars in either Group 1 or Group 2. If the selected bar in Group 3 weighs equally as the selected bar in either Group 1 or Group 2, then you can safely assume that the other bar (Bar F in this case) is the fake gold bar. If the selected bar in Group 3 weighs differently than the selected gold bar in either Group 1 or Group 2, then you can safely assume that the selected bar in Group 3 that you just weighed is the fake gold bar (in this case Bar E).

In the slider below, please indicate the **minimum royalty amount you are willing to accept** to license this solution.

0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5

In Dollars (\$)

Next >

Figure 47: Provider Medium Quality Algorithm Price Choice

According to the expert, this solution has a **50% chance of success**.

First, weigh gold A and B against gold C and D. If blocks A and B are equal to blocks C and D, then gold E or F must be fake. To find which one, E or F, is fake, you must replace gold A with gold E. If they are still equal, F is the fake gold. If they are no longer equal, E is the fake gold.

If after comparing the weights of A and B against C and D and finding that the two weights are not equal, you know that blocks E and F are real. You must then weigh gold A against gold B. If they are equal, then you know C or D is fake. On your third and last time to weigh the blocks you would replace block A with block C. If now the two blocks (C and B) are unequal, you know that C is the fake block. If they are still equal then you know that D is fake.

If you weigh A against B and they are not equal, you must replace A with C. If they are still unequal, B is fake. If they are now equal, then A was fake.

In the slider below, please indicate the **minimum royalty amount you are willing to accept** to license this solution.

00.511.522.533.544.55

In Dollars (\$)

Next >

Figure 48: Provider Low Quality Algorithm Price Choice

According to the expert, this solution has a **less than 20% chance of success**.

In order to figure out which bar is fake gold, I think a good solution would be to first start with by weighing three and three. Since all the real gold bars weigh the same then you could see which bunch of three produces a different weight. By doing this, you can see which side has the fake gold mixed in with the real ones. After you do this, you can take the three that are different weights and put two on one side and one on the other. Then, you can switch the order and swap out the one with one of the bars from the side with two. By doing this, you can determine a difference in weight. Two must weigh the same and one must weigh differently. Isolating first to three and then doing two and one twice will help determine the fake gold bar.

In the slider below, please indicate the **minimum royalty amount you are willing to accept** to license this solution.

00.511.522.533.544.55

In Dollars (\$)

Next >

Figure 49: User High Quality Algorithm Price Choice

According to the expert, this solution has a **80% or greater chance of success.**

Pair one gold bar with another (no repeats) in order to make 3 separate groups of gold bars with 2 gold bars in each group.
Ex.
Group 1: A and B
Group 2: C and D
Group 3: E and F

Weigh Group 1 against Group 2. You should expect 1 of 2 results. Either the group of bars equally weigh each other or not. Note this result.
Then Weigh Group 1 against Group 3. Again, you should expect 1 of 2 results. Either the group of bars should weigh each other or not. Note this result.

Based on your results, this is the solution and follow the description:

If Group 1 and Group 2 did NOT weigh the same AND Group 1 and Group 3 did NOT weigh the same, then you know that the one of the gold bars in Group 1 is fake. Choose either bar A or bar B and then weigh it against one of the gold bars in either Group 2 or Group 3. If the selected bar (let's say Bar B) in Group 1 weighs equally as the selected bar in either Group 2 or Group 3, then you can safely assume that the other bar (Bar A in this case) is the fake gold bar. If the selected bar in Group 1 weighs differently than the selected gold bar in either Group 2 or Group 3, then you can safely assume that the selected bar in Group 1 that you just weighed is the fake gold bar (in this case Bar B).

If Group 1 and Group 2 did NOT weigh the same AND Group 1 and Group 3 did weigh the same, then you know that the one of the gold bars in Group 2 is fake. Choose either bar C or bar D in Group 2 (let's say Bar C) and then weigh it against one of the gold bars in either Group 1 or Group 3. If the selected bar in Group 2 weighs equally as the selected bar in either Group 1 or Group 3, then you can safely assume that the other bar (Bar D in this case) is the fake gold bar. If the selected bar in Group 2 weighs differently than the selected gold bar in either Group 1 or Group 3, then you can safely assume that the selected bar in Group 2 that you just weighed is the fake gold bar (in this case Bar C).

If Group 1 and Group 2 did weigh the same AND Group 1 and Group 3 did NOT weigh the same, then you know that the one of the gold bars in Group 3 is fake. Choose either bar E or bar F in Group 3 (let's say Bar E) and then weigh it against one of the gold bars in either Group 1 or Group 2. If the selected bar in Group 3 weighs equally as the selected bar in either Group 1 or Group 2, then you can safely assume that the other bar (Bar F in this case) is the fake gold bar. If the selected bar in Group 3 weighs differently than the selected gold bar in either Group 1 or Group 2, then you can safely assume that the selected bar in Group 3 that you just weighed is the fake gold bar (in this case Bar E).

In the slider below, please indicate the **maximum royalty amount you are willing to pay** to license this solution from its owner.

0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5

In Dollars (\$)

Next >

Figure 50: User Medium Quality Algorithm Price Choice

According to the expert, this solution has a **50% chance of success**.

First, weigh gold A and B against gold C and D. If blocks A and B are equal to blocks C and D, then gold E or F must be fake. To find which one, E or F, is fake, you must replace gold A with gold E. If they are still equal, F is the fake gold. If they are no longer equal, E is the fake gold.

If after comparing the weights of A and B against C and D and finding that the two weights are not equal, you know that blocks E and F are real. You must then weigh gold A against gold B. If they are equal, then you know C or D is fake. On your third and last time to weigh the blocks you would replace block A with block C. If now the two blocks (C and B) are unequal, you know that C is the fake block. If they are still equal then you know that D is fake.

If you weigh A against B and they are not equal, you must replace A with C. If they are still unequal, B is fake. If they are now equal, then A was fake.

In the slider below, please indicate the **maximum royalty amount you are willing to pay** to license this solution from its owner.

0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5

In Dollars (\$)

Next >

Figure 51: User Low Quality Algorithm Price Choice

According to the expert, this solution has a **less than 20% chance of success**.

In order to figure out which bar is fake gold, I think a good solution would be to first start with by weighing three and three. Since all the real gold bars weigh the same then you could see which bunch of three produces a different weight. By doing this, you can see which side has the fake gold mixed in with the real ones. After you do this, you can take the three that are different weights and put two on one side and one on the other. Then, you can switch the order and swap out the one with one of the bars from the side with two. By doing this, you can determine a difference in weight. Two must weigh the same and one must weigh differently. Isolating first to three and then doing two and one twice will help determine the fake gold bar

In the slider below, please indicate the **maximum royalty amount you are willing to pay** to license this solution to others.

0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5

In Dollars (\$)

Next >

2.4 Results

Figure 52: Provider Pre-Results Page

Before the next page, the computer will randomly select

- One of your three solutions
- A royalty amount between \$0 and \$5 for that solution.

Next >

Figure 53: Provider Results Example 1

The solution with a greater than **80%** chance of success has been randomly selected for you to license. The royalty amount generated by the computer is **\$1.11**.

Next >

Figure 54: Provider Results Example 2

The solution with a **50%** chance of success has been randomly selected for you to license. The royalty amount generated by the computer is **\$3.01**.

Next >

Figure 55: User Pre-Results Page

Before the next page, the computer will randomly select

- One of the three solutions
- A royalty amount between \$0 and \$5 for that solution.

Next >

Figure 56: User Results Example 1

The solution with a greater than **80%** chance of success has been randomly selected for you to license. The royalty amount generated by the computer is **\$1.99**.

Next >

Figure 57: User Results Example 2

The solution with a greater than **80%** chance of success has been randomly selected for you to license. The royalty amount generated by the computer is **\$3.89**.

Next >

2.5 End

Figure 58: End Page

You will receive an electronic Amazon gift card with your total earnings (\$3 plus any additional earnings from this survey). If you win the \$10 lottery, it will also be added to your Amazon gift card.

Please enter the email address you would like to have your Amazon gift card delivered to.

[OPTIONAL] The researcher would like to hear your feedback, comments, and suggestions based on your experience with this survey. If you have any thoughts, please include them in the comment box below:

Thank you for your contribution to research in economics!
Please click "Next" to submit your survey response.

Next >